

Dynamical Systems That Sort Lists, Diagonalize Matrices, and Solve Linear Programming Problems

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ABSTRACT

We study a class of smooth, continuous-time dynamical systems defined on the space of real symmetric matrices. The principal equation of interest is the double-bracket commutator flow: $dH/dt = [H, [H, N]]$, where $H(t)$ and N are symmetric $n \times n$ matrices and $[A, B] = AB - BA$ is the Lie bracket. We prove that this differential equation defines an isospectral gradient flow on the adjoint orbit of the orthogonal Lie group $O(n)$. When N is chosen as a fixed diagonal matrix with distinct entries in strictly ascending order, the steady-state solution $H(\infty)$ is diagonal, with its diagonal entries arranged in strictly ascending order matching N . Consequently, the flow sorts arbitrary lists of numbers, computes matrix eigenvalues, and solves linear programming problems in continuous physical time.

1. Introduction and Motivation

Traditional algorithms in computer science treat sorting as an inherently discrete, combinatorial task requiring sequential comparisons and swaps. However, in analog computing, Hamiltonian mechanics, and Riemannian geometry, optimization problems frequently admit natural representations as continuous dynamical systems.

In this work, we demonstrate that sorting an unordered list of real numbers can be realized as the natural evolution of an autonomous gradient flow on the homogeneous manifold of symmetric matrices under the adjoint action of $SO(n)$.

2. Lie Bracket Formulation & Geometric Structure

Let $S(n)$ denote the vector space of $n \times n$ real symmetric matrices, and let $so(n)$ denote the Lie algebra of skew-symmetric matrices. Equip $S(n)$ with the canonical Frobenius trace inner product $\langle A, B \rangle = \text{Tr}(AB)$.

Given an initial symmetric matrix $H(0)$ whose eigenvalues are the numbers to be sorted, its adjoint orbit is $O(H(0)) = \{ Q H(0) Q^T : Q \in SO(n) \}$. Every matrix along this orbit shares identical eigenvalues with $H(0)$. Thus, motion along this orbit is strictly isospectral.

3. The Double-Bracket Equation

The evolution of $H(t)$ is governed by:

$$dH/dt = [H, [H, N]] = [[N, H], H]$$

Here $[A, B] = AB - BA$. Observe that $\Omega = [N, H]$ is skew-symmetric, since $\Omega^T = (NH - HN)^T = HN - NH = -\Omega$. Thus, $[H, [H, N]] = [H, -\Omega] = [\Omega, H]$ is the commutator of a skew-symmetric matrix with H .

By Jacobi's identity and derivative of the matrix exponential, $H(t) = Q(t) H(0) Q(t)^T$ where $dQ/dt = \Omega(t) Q(t)$.

This guarantees that the spectrum of $H(t)$ is strictly conserved for all $t \geq 0$.

4. Gradient Ascent of the Rearrangement Potential

Consider the potential function $\Phi(H) = \text{Tr}(HN) = \sum_{i,j} H_{ij} N_{ji}$. Brockett proved that the double-bracket flow is precisely the Riemannian gradient ascent flow of $\Phi(H)$ restricted to the compact orbit $O(H(0))$:

$$\text{grad } \Phi(H) = [H, [H, N]]$$

Computing the time derivative of $\Phi(H)$:

$$\frac{d}{dt} \Phi(H) = \text{Tr} \left(\left(\frac{dH}{dt} \right) N \right) = \text{Tr} \left([H, [H, N]] N \right) = -\| [H, N] \|_F^2 \leq 0$$

Therefore, $\Phi(H)$ increases strictly monotonically along all non-equilibrium trajectories. Since the orbit is compact and Φ is bounded, the system must converge to an equilibrium point where $\| [H, N] \|_F = 0$.

5. Sorting Equilibrium and Global Attractor

At equilibrium, $[H(\infty), N] = 0$. Because $N = \text{diag}(1, 2, \dots, n)$ has distinct eigenvalues, any symmetric matrix commuting with N must itself be diagonal.

By the classical Rearrangement Inequality, $\text{Tr}(HN) = \sum_{i=1}^n H_{ii} \cdot i$ attains its unique global maximum over all permutations of eigenvalues if and only if $H_{11} < H_{22} < \dots < H_{nn}$. Hence, the unique asymptotically stable attractor of the flow is the sorted diagonal matrix!